Homework 1

Stat 9340

Due: Feb 9 2023

1)

Say we have (x, y) data generated according to the process

$$y_i = \theta_1 \left(\sin(x) + \cos(\theta_2 x)\right) + \epsilon_i$$

for $i = 1, \ldots, n$ and $\epsilon_i \sim N\left(0, \sigma^2\right)$. You will find the data on Canvas in the file `HW1_Problem1_Data.csv`.

a. Read the data into the software of your choice and plot y vs x.

In the remaining exercises for this problem, you should assume a squared error loss objective function and code the algorithms yourself as well compute gradients "by hand" and hardcode their evaluation in your program.

b. Code the SGD optimizer and use it to estimate θ_1 and θ_2 above. Experiment with various starting values, learning rates, and batch sizes. Study the number of iterations it takes to reach convergence and how well the resulting estimates fit the data.

c. Add a momentum term to your SGD code and study convergence/estimation again while also varying the momentum. How does this compare to your basic SGD?

d. Now code the Adam algorithm and perform a similar analysis with it's tuning parameters and compare to the above.

e. Code a PSO algorithm to minimize the loss and compare its performance to the other methods. Be sure to experiment with different swarm sizes and tuning parameters.

f. Include a few plots of the estimated fits for parameter estimates obtained from optimizers that performed well and not so well.

2)

The 2-dimensional *Rosenbrock function* is given by

$$f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$$

This function is typically used to evaluate the performance of optimization algorithms. It has a parabolic valley, which is fairly easy to converge into, but finding the global minimum of $(1, 1)$ can be a challenge.

a. Make a contour plot of the Rosenbrock function.

b. Compute the gradient of the Rosenbrock function and display gradient vectors on the contour plot from *a*.

c. With a starting value of $(-0.1, 2.5)$, use SGD, SGD with momentum, and Adam to attempt to arrive at the minimum location of $(1, 1)$ via Tensorflow's optimization routines. Store the intermittent values of x_1 and x_2 and trace the path of the optimizer as it looks for the global minimum on your plot from b.

d. Experiment with 2 more optimizers or optimizer configurations of your choice: Documentation Link

e. The N-dimensional Rosenbrock function is given by

$$f(\mathbf{x}) = \sum_{i=1}^{N-1} \left[100\left(x_{i+1} - x_i^2\right)^2 + (1 - x_i)^2\right] \quad \text{where } \; N \geq 2 \text{ and } \; \mathbf{x} = (x_1, \ldots, x_N) \in \mathbb{R}^N.$$

Study the effects of the dimension of the function on the difficulty of achieving convergence to the global minimum of $\mathbf{1}_N$ using various optimizers. Consider criteria such as number of iterations it takes to converge, how often each algorithm converges to the global minimum, or anything else you can think of. A small amount of extra credit will be awarded if your program is parameterized by the dimension of the function.

3

Use symbolic differentiation on the 2-dimensional Rosenbrock function from problem 2 to numerically evaluate the gradient $\left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}\right)$ at $x_1 = x_2 = 0$. Then construct the evaluation trace and perform forward and reverse mode AD by hand to evaluate the two partial derivatives at that same point. You do not need to draw the graph in your submission but should show the numerical values in a table as well as well as the construction of the intermediate variables v_i.